GLOBE SCOTT MOTORS · FREE ENGINEERING RESOURCE

Discontinued Motor Replacement Data Capture Sheet

For maintenance engineers and OEM designers replacing an obsolete DC or geared motor

Complete Sections 1 to 5 on the old motor before it is removed or scrapped, then use Section 6 and the acceptance table to approve the replacement. Keep the completed sheet with the machine file as the specification for future spares.

Step 1 — Nameplate record

- ☐ Photograph the nameplate in raking light and file the image with this sheet
- ☐ Record voltage (V), output (W), speed (rpm) and current (A) exactly as printed
- ☐ Record duty type: S1, S2 with time limit, or S3 with cyclic duration factor (%)
- ☐ Record insulation class, IP code and rated ambient (°C)
- ☐ Record part number, serial number, date code and the machine it came from
- ☐ Note that plate speed is the loaded speed at rated torque, not the no-load speed

Step 2 — Electrical bench tests on the old motor

- ☐ Decouple the motor and run at plate voltage from a stable supply for 5 minutes
- ☐ Record no-load speed N0 (rpm) with an optical tachometer
- ☐ Record no-load current I0 (A) with a true-DC clamp meter or shunt
- ☐ Measure terminal resistance Rt by locked-rotor low-voltage test at about rated current; average 3 rotor positions; note temperature
- ☐ Calculate w0 = N0 × 0.1047 rad/s, then Ke = (V – I0 × Rt) / w0; Kt in Nm/A equals Ke in V·s/rad
- ☐ Record rotation viewed from the drive end with positive on the marked lead, and lead colours
- ☐ Cross-check: speed predicted at plate current with a warm winding should be close to plate rpm

Step 3 — In-service load measurement

- ☐ Log terminal voltage at the motor (not at the supply), current and speed over a full machine cycle
- ☐ Capture start, steady running, heaviest product, cold start, jam and reversal events
- ☐ Convert each current reading to torque: T = Kt × (I – I0)
- ☐ Calculate RMS torque over the cycle for heating and note the peak torque separately
- ☐ Record ambient temperature at the motor and frame temperature after 1 hour of normal duty
- ☐ Record on-time, off-time and starts per hour to confirm the real duty type

Step 4 — Mechanical interface

- ☐ Mounting type: B3 foot, B5 flange or B14 face
- ☐ Spigot (register) diameter by micrometer and spigot depth by depth gauge
- ☐ Bolt PCD, hole count, clearance hole size or thread size
- ☐ Shaft diameter at two positions to 0.01 mm, extension length from the mounting face
- ☐ Key width × height × length, or across-flat dimension and flat length
- ☐ Overall length and maximum diameter, plus installed clearance including connector and cable bend

- [ ] Lead exit clock position relative to mounting holes, cable length, connector part number and pin-out

Step 5 — Accessories and controller

- [ ] Brake: coil voltage, holding torque, supply source, manual release
- [ ] Encoder or Hall sensor: PPR, channels, output type, supply voltage
- [ ] Gearbox: ratio by turn count over 5 or more output turns, output shaft, back-drive behaviour
- [ ] Controller: current limit setting, fuse or breaker rating, thermal protector wiring
- [ ] EMI suppression components fitted at the motor terminals

Step 6 — Validation and release of the replacement

- [ ] Bench: no-load speed, current, resistance and rotation against Step 2 and the acceptance table
- [ ] Fit: spigot engagement, connector mating and shaft runout at the extension (typically 0.05 mm or better)
- [ ] Operating point: current within ±10 % of the logged value at the same machine load
- [ ] Thermal: run until frame temperature changes by less than about 2 K per hour; compare rise with the old motor
- [ ] Fault cases: deliberate jam with real protection fitted, brake hold and release, encoder direction
- [ ] Endurance: pilot machine for an agreed period before converting the fleet
- [ ] Issue a performance specification, interface drawing and retained sample for future supply

Typical acceptance targets for a drop-in replacement

Parameter	Method	Typical target
No-load speed	Tachometer at plate voltage, decoupled	Within ±5 % of old motor
Back-EMF constant Ke	Calculated from no-load test	Within ±5 %
Terminal resistance Rt	Locked-rotor test, 3 positions	Within ±15 %
Current at machine load	True-DC clamp at the same load	Within ±10 % of logged value
Rotation and polarity	Viewed from drive end	Identical
Spigot and PCD	Micrometer, centre distance	Same fit; PCD within ±0.1 mm
Shaft extension	Depth gauge from mounting face	Within ±0.5 mm unless position-critical
Brake and encoder	Data plate and signal check	Same coil voltage and signal type

Targets are typical engineering starting points for general industrial drives. Tighten them for synchronised, position-critical or high-speed machines, and always confirm against the machine builder's requirements.

Need help sizing or replacing a motor?
Globe Scott Motors has manufactured electric motors in Gujarat since 1985 and today builds PMDC, BLDC and geared motors, PMDC transaxles and railway signalling motors. Share your load, duty cycle and envelope at www.globescott.net/contact-us/

General engineering guidance only. Always follow the equipment manufacturer's instructions, applicable standards and site safety rules (isolation and lock-out/tag-out) before working on electrical or rotating equipment.